

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>1</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |
|          | <p><b>(a)</b> units of <math>T</math>: s, <math>l</math>: m and <math>M</math>: kg</p> <p><math>K^2 = T^2 E / M l^3</math><br/> hence units: <math>s^2 \text{kg m}^{-1} \text{s}^{-2} / \text{kg}^3 (= \text{m}^{-4})</math> C1<br/> (for correct substitution of units of <math>T</math>, <math>l</math> and <math>M</math>,<br/> allow omission of square/square root but do not allow e.g. <math>\text{s}^{-1}</math>)</p> <p>units of <math>K</math>: <math>\text{m}^{-2}</math> A1</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | [2] |
|          | <p><b>(b)</b> <math>E = [(1.48 \times 10^5)^2 \times 0.2068 \times (0.892)^3] / (0.45)^2</math><br/> <math>= 1.588 \times 10^{10}</math> B1</p> <p>% uncertainty in <math>E = 4\% + 0.6\% + 0.1\% + 3\%</math><br/> <math>= 7.7\%</math> C1<br/> 7.7% of <math>E = 1.22 \times 10^9</math> (allow ecf) A1<br/> OR<br/> <math>E_{\text{max}} = [(1.48 \times 10^5) (1.015)]^2 [0.2068 (1.001)] [0.892 (1.002)]^3 / [0.45 (0.980)]^2</math><br/> <math>= (150220)^2 (0.2070068) (0.893784)^3 / (0.441)^2</math><br/> <math>= 1.715 \times 10^{10}</math><br/> <math>E_{\text{min}} = [(1.48 \times 10^5) (0.985)]^2 [0.2068 (0.999)] [0.892 (0.998)]^3 / [0.45 (1.020)]^2</math><br/> <math>= (145780)^2 (0.2065932) (0.890216)^3 / (0.459)^2</math><br/> <math>= 1.470 \times 10^{10}</math> C1 (<math>E_{\text{max}}</math> or <math>E_{\text{min}}</math> correct)<br/> <math>\Delta E = \frac{1}{2} (E_{\text{max}} - E_{\text{min}})</math> or <math>E_{\text{max}} - E</math> or <math>E - E_{\text{min}}</math> A1 (<math>= 1.22 \times 10^9</math>)</p> <p><math>E = (1.6 \pm 0.1) \times 10^{10} \text{ kg m}^{-1} \text{s}^{-2}</math> (allow ecf) B1<br/> (award this mark if absolute uncertainty is in 1 sf and actual value follows the dp of the uncertainty)</p> | [4] |

|          |            |                                                  |    |
|----------|------------|--------------------------------------------------|----|
| <b>2</b> | <b>(a)</b> | <b>(i)</b> product of mass and (linear) velocity | B1 |
|----------|------------|--------------------------------------------------|----|